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Electrical conductivity and dynamics of quasi-solidified lithium-ion conducting ionic liquid at oxide particle surfaces

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ABSTRACT

Lithium ion conducting solid-state composites consisting of lithium ion conducting ionic liquid, lithium bis (trifluoromethanesulfonyl)amide (Li-TFSA) dissolved 1-ethyl-3-methyl imidazolium bis(trifluoromethylsulfonyl) amide (EMI-TFSA), denoted by [yMLi$^+$][EMI$^+$][TFSA$^-$] in this study, and various oxide particles such as SiO_2, Al_2O_3, TiO_2 (anatase and rutile) and 3YSZ are synthesized via a liquid route for the molar concentration of lithium, y, to be 1. The composite consists of SiO_2 and the ionic liquid with $y=0.2$ was also prepared. The ionic liquid are quasi-solidified at the above oxide particle surfaces when x is below 40 for $y=1$ and x is below 30 for $y=0.2$, corresponding to the confinable thickness of the ionic liquid at the oxides' surfaces to be approximately 5–10 nm regardless of the oxide compositions. The electrical conductivities of x vol.%[yMLi$^+$][EMI$^+$][TFSA$^-$]–SiO_2, Al_2O_3, TiO_2s or 3YSZ composites are evaluated by ac impedance measurements. The quasi-solid-state composites exhibited liquid-like high apparent conductivity, e.g. $10^{-3.3}$–$10^{-2.0}$ S cm^{-1} in the temperature range of 323–538 K for SiO_2–ionic liquid composites with $y=1$. The self-diffusion coefficients of the constituent species of x vol.% [yMLi$^+$][EMI$^+$][TFSA$^-$] (x is below 40, $y=0.2$ and 1) $-SiO_2$ are evaluated by the Pulse Gradient Spin Echo (PGSE)-NMR technique in the temperature range of 298–348 K. By the quasi-solidification of the ionic liquid at SiO_2 particle surfaces, the absolute values of the diffusion coefficients of all constituent species decreased. The SiO_2 surfaces work to promote ionization of ion pair, [EMI$^+$][TFSA$^-$], while significant influence on the solvation coordination, $[Li(TFSA)_{n+1}]^{n-}$, was not observed. The apparent transport numbers of Li-containing species both in the bulk and the quasi-solidified ionic liquid showed similar values with each other, which was evaluated to be in the range of 0.010–0.017 for $y=0.2$ and 0.051–0.093 for $y=1$ in the abovementioned temperature range.

1. Introduction

In order to achieve high energy density, safety and stable large-scale secondary battery, it is necessary to develop highly lithium conductive solids for use in all solid lithium secondary battery and lithium–air secondary battery as electrolytes. To date, the electrical conductivity of the various solids were investigated so far, the number of materials having sufficient lithium ion conductivity for the above applications are limited [1–3]. On the other hand, ionic liquid is one of the promising candidates as electrolyte materials in electrochemical devices. This is because ionic liquids have high ion conductivity and wide potential window. In addition, they are thermally and chemically stabile, involatile and nonflammable.

In recent years, it was revealed that ionic liquids are quasi-solidified at the oxide particle surfaces attributed to the strong interaction between them. The composite materials consisting of the ionic liquid and oxide composites keep liquid-like high electrical conductivity [4–8]. The structure and physical property in solidified ionic liquid are reported to deviate from those in the bulk [9–11]. Likewise, arguments are on the dynamics of ionic liquids in the quasi-solidified layer at the oxide surfaces. Mezger et al. have modeled the structure based on the high energy X-ray reflectivity measurement. They revealed that the ions in ionic liquid have highly ordered structure at the surface of charged sapphire substrate [12]. Ueno et al. have investigated the viscosity near the interface consisting of ionic liquid and SiO_2. They revealed that the viscosity exhibits higher values approximately three orders of magnitude just beneath the oxide surface than the bulk. It gradually relaxed within 10 nm from the silica surface [13]. Since the electrical conductivity of ionic liquid has a positive dependence on inverse of viscosity [14–16], the influence of the high ordering and/or increase of the viscosity are considered to affect the dynamics of ionic liquids.

NMR spectroscopic studies were demonstrated for quasi-solidified ionic liquid at silica surfaces [6,7,17,18]. The broadening of ^{1}H NMR spectra was observed by confinement of ionic liquid at silica particles [17], suggesting that the motion of ionic liquid becomes slow. Some researchers suggest that the motion of ionic species has negligible

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effect on motion of ionic species [5–8]. However, Shimano et al. has pointed out that the temperature dependence of the apparent electrical conductivity became large when the mixing ratio of ionic liquid to oxides decreased, i.e. decrease in thickness of ionic liquid at oxide particle surfaces [4]. Since the temperature dependence of the electrical conductivity reflects the dynamics natures, the difference of the temperature dependence of the apparent conductivity between the ionic liquid quasi-solidified composite and the ionic liquid bulk infers the change in the transport properties of ionic species.

Ueno et al. have investigated the effect of SiO_2 dispersion on the diffusivities of constituent ionic species in ion gels such as EMI^+ and $TFSA^-$. They have found that the small amount of SiO_2 addition decrease of both anions and cations. The diffusion coefficient decreases with increasing SiO_2 content [18]. Saito and her workers have investigated the diffusion coefficients of the ionic species in SiO_2-dispersed ion gel consisting of Li-TFSA dissolved 1-butyl-2,3-dimetylimidazolium bis(trifluoromethylsulfonyl)amide (BDMI-TFSA), $[Li^+][BDMI^+][TFSA^-]$. Introduction of SiO_2 particle enhanced the diffusion coefficient of Li^+ by approximately 0.4 orders of magnitude. The diffusion coefficient of $BDMI^+$ decreased while that of $TFSA^-$ was not affected [19]. The effect of the oxide on the dynamics of the constituent ionic species in the quasi-solidified ionic liquid at oxide surface might be more significant than oxide dispersed ion gel because the former has higher surface density than the latter.

In order to achieve better understanding of the ion dynamics in quasi-solidified ionic liquid at oxide particle surfaces, we systematically fabricated ionic liquid quasi-solidified composite materials consisting of lithium ion conducting ionic liquids, $[yMLi^+][EMI^+][TFSA^-]$, and various oxide particles such as SiO_2, Al_2O_3, TiO_2 (rutile and anatase) and 3YSZ for the molar concentration of lithium, y, to be 1 and SiO_2 for $y = 0.2$ as a function of ionic liquid volume ratios. The state diagram of oxide–ionic liquid composite is proposed based on the appearance of the composite materials. The ion dynamics, especially how SiO_2 surface affects the dynamics in the quasi-solidified ionic liquid, was discussed based on the results of the electrical conductivity measurements and the results of the NMR spectroscopic analysis.

2. Experimental

2.1. Materials

The oxides such as SiO_2, Al_2O_3, TiO_2 (rutile and anatase) and 3YSZ are used as a matrix. Detailed information on the oxides is provided in Table 1. For TiO_2 (rutile), the grains are crushed by a planetary ball mill (Fritsch, P-5) using φ 2 mm ziroconia balls for 3 h at 400 rpm, and then, φ 0.6 mm zirconia balls for 15 h at 400 rpm in zirconia pot filled with ethanol. Powder of Li-TFSA (Kishida Chemical Co., Ltd. or Kanto Chemical Co., Inc.) was dissolved into EMI-TFSA (Wako Pure Chemical Industries, Ltd. or Kanto Chemical Co., Inc., the ion structures are shown in Fig. 1) to be 1 M and 0.2 M of lithium concentrations. The solutions were separately mixed together with various oxide particles for $y = 1$ and SiO_2 for $y = 0.2$ to be the ionic liquid volume ratio of 0.025–0.5 in organic solvents such as *N*-methylpyrrolidone (NMP, Wako Pure Chemical Industries, Ltd.) or methanol (Wako Pure Chemical Industries, Ltd.). The mixture was stirred for at least 3 h, then, the solvents were evaporated by a hot plate. Microstructure observation and composition analysis were performed using a scanning electron microscope (SEM, JSM-7001 F, JEOL) and an energy dispersive X-ray spectrometer (EDS, Inca x-act, Oxford Instruments), respectively.

Table 1
List of oxides used in this study.

Composition	Supplier	Relative surface areas/ $m^2\,g^{-1}$
SiO_2	Nippon Shokubai Co., Ltd.	54.8
Al_2O_3	Taimei Chemical Co., Ltd.	13.0
TiO_2 (anatase)[a]	Nippon Aerosil®	49.2
TiO_2 (rutile)	Kojundo Chemical Laboratory Co., Ltd.	49.1
3YSZ	Tosoh Corp.	17.0

[a] The oxide is a mixture of 80% anatase and 20% rutile.

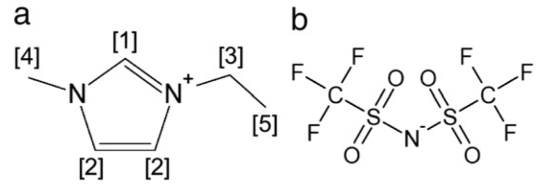

Fig. 1. Ion structures of (a) EMI^+ and (b) $TFSA^-$.

2.2. NMR spectroscopic measurements

^1H, ^{19}F and ^7Li static and MAS-NMR spectra of SiO_2– and Al_2O_3–$[1MLi^+][EMI^+][TFSA^-]$ composites were obtained by Bruker Avance 600 spectrometer (Bruker BIOSPIN) at a magnetic field of 14.1 T with 2.5 mm φ or 4 mm φ MAS probe. The powders of adamantane, LiCl and 1-fluoroadamantane were used as the chemical shift standards of ^1H (1.91 ppm), ^{19}F (0 ppm) and ^7Li (0 ppm), respectively. The measurements were conducted with a spinning rate of 10 kHz at room temperature. Pulse Gradient Spin Echo (PGSE)-NMR measurements for the nuclei, ^1H, ^7Li and ^{19}F, were performed to determine the diffusion coefficients of the constituent species in the temperature range of 298–348 K. Bruker Avance 400 spectrometer (Bruker BIOSPIN), equipped with a Diff60 diffusion probe (Bruker BIOSPIN), having a magnetic field of 9.4 T, was employed. H_2O (4.877 ppm), 1 M $LiCl/H_2O$ (0 ppm) and C_6H_5F (0 ppm) were used as chemical shift standards of ^1H, ^7Li and ^{19}F, respectively. The self-diffusion coefficients were determined by the Hahn's spin-echo pulse sequence. The magnetic field was calibrated by the above standards at 298 K based on the literature values [20].

2.3. Electrical conductivity measurements

The electrical conductivity of ionic liquid, $[yMLi^+][EMI^+][TFSA^-]$, and the apparent conductivities of the uniaxially pressed body of the oxide–ionic liquid composites were evaluated by a two-probe ac technique with using platinum plates as electrodes in argon flow. Platinum lead wires are connected to a set of an electro-analytical system consisting of a potentiostat/galvanostat (Toho Technical Research Co., Ltd., PS2000) and a frequency response analyzer (NF Corp., FRA5087). The temperature dependence of the electrical conductivities was evaluated in the temperature range of 298–538 K.

3. Results and discussion

3.1. State of ionic liquid in composite

Fig. 2(a) and (b) shows the typical photographs of SiO_2–x vol.% $[1MLi^+][EMI^+][TFSA^-]$ ($x = 40$ and 50). The mixtures of SiO_2 and ionic liquid was a white powder, i.e. quasi-solid-state, for $x = 40$ while it became a gel when $x = 50$. In the case of $y = 0.2$, the ionic liquid was a quasi-solidified with $x = 30$ while that was a transparent gel with $x = 40$. For the other oxides such as 3YSZ, TiO_2s, ionic liquids could be solidified when x is below 40.

SEM images and distributions of silicon, oxygen, carbon and fluorine typically for SiO_2–x vol.%$[1MLi^+][EMI^+][TFSA^-]$ mixtures are shown in Fig. 3(a)–(d) are the ones for $x = 0$, 2.5, 40 and 50, respectively. Fluorine distribution comes from $TFSA^-$ (see Fig. 1). Since the specimens are mounted on a carbon tape for SEM and EDX analyses, carbon is also distributed at the analyzed areas where the composite powders do not exist.

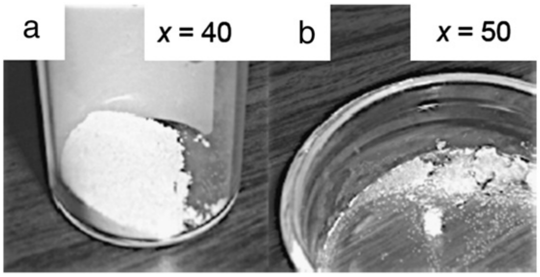

Fig. 2. Photographs of SiO_2–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] composites for (a) $x = 40$ and (b) $x = 50$.

It was found from the SEM images that the SiO_2 grains are agglomerated as ionic liquid fraction increased. The SiO_2 particles are connected attributed to strong interaction between ionic liquids and oxide surfaces in the composites. The more the composite has ionic liquids, the easier the ionic liquid phase captures the surrounding SiO_2 particles. It was found from EDS mapping that Si and F coexist together when x is below 40. On the other hand, the above elements were homogeneously distributed in analyzed areas for $x = 50$. Those suggest that ionic liquid is dispersed at oxide particle surfaces when x is below 40 vol.% while the oxide particles are dispersed in ionic liquid when x is above 50 vol.%. This implies that the boundary dividing quasi-solid-state and gel exists between the above ionic liquid volume ratios.

In order to semi-quantitatively investigate the confinable ionic liquid thickness as a function of the various oxide particles, the authors

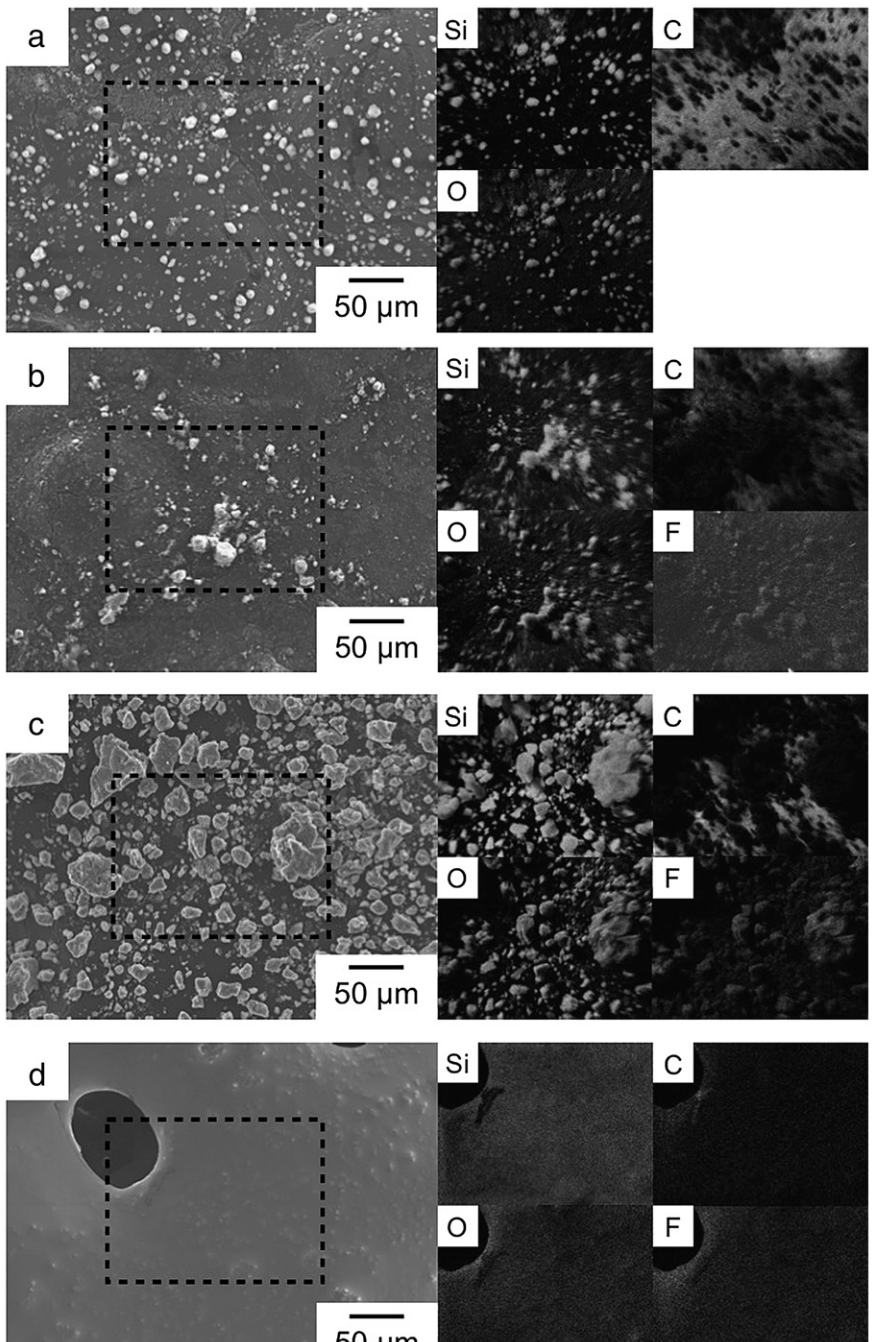

Fig. 3. SEM image and Si, O, C and F distributions of SiO_2–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] composites: (a) $x = 0$, (b) $x = 2.5$, (c) $x = 40$ and (d) $x = 50$.

adopted the following state diagram. Since the density of oxides varies depending on the oxide composition, simple comparison only by the relative surface areas is meaningless. The weight, the density and the relative surface areas of the oxides are defined to be w_s [g], D_s [g cm^{-3}] and A_r [m^2 g^{-1}], respectively. The weight, the density, the average thickness and volume ratio of ionic liquid layers are defined as w_{IL} [g], D_{IL} [g cm^{-3}], $t_{ave.}$ [m] and V_{IL}, respectively. X-axis and Y-axis are defined by,

$$X = D_s A_r \tag{1}$$

$$Y = V_{IL} = \frac{w_{IL}}{D_{IL}} \frac{1}{\frac{w_{IL}}{D_{IL}} + \frac{w_s}{D_s}} = \frac{w_{IL} D_s}{w_{IL} D_s + w_s D_{IL}}. \tag{2}$$

Consistency in ionic liquid volume provides,

$$V_{IL} = t_{ave.} w_s A_r = \frac{w_{IL}}{D_{IL}} 10^{-6}. \tag{3}$$

Solving for w_s of Eq. (3) and substitution into Eq. (2) leads to,

$$Y = \frac{t_{ave.} A_r D_s}{t_{ave.} A_r D_s + 10^{-6}}. \tag{4}$$

Substitution of Eq. (1) into Eq. (4) gives,

$$Y = \frac{t_{ave.} X}{t_{ave.} X + 10^{-6}}. \tag{5}$$

By using Eq. (5), the average confinable thickness of ionic liquid, $t_{ave.}$, can be compared as a function of oxides having different densities. Fig. 4 shows the state diagram of the ionic liquid in the ionic liquid–oxide composites. The bars in the figures are drawn by tying the ionic liquid volume ratios between quasi-solidified and gel/liquid states. This means the boundary dividing the ionic liquid quasi-solidified and gel/liquid states exists on the bars. The literature data provided by Shimano et al. [4] are also shown in the figure. The solid curve in the figure represents the X–Y relation assuming $t_{ave.}$ to be 1–20 nm. It should be noted here that the surface state, ionic liquid composition, grain size distributions etc. are the possible factors determining the ionic liquid confining capability. In addition, we need to assume that the ionic liquid thickness at oxide surfaces is homogeneous. Thus, the simplified determination of confined ionic liquid thickness from Eq. (5) is absolutely semi-quantitative. However, the interesting fact from the figure is that the confinable ionic liquid thickness is approximately 5–10 nm regardless of oxide and ionic liquid compositions except for SiO_2-1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)amide (BMI-TFSA) composite with $D_s A_r = 0.022$. This value is roughly in consistent with the report by Liu et al. who reported that the confinable ionic liquid at nano-SiO_x particles is approximately 20 nm [10]. In addition, we also saw similarity to the distance which the viscosity differs from the bulk as reported by Ueno et al., i.e. approximately 10 nm [13].

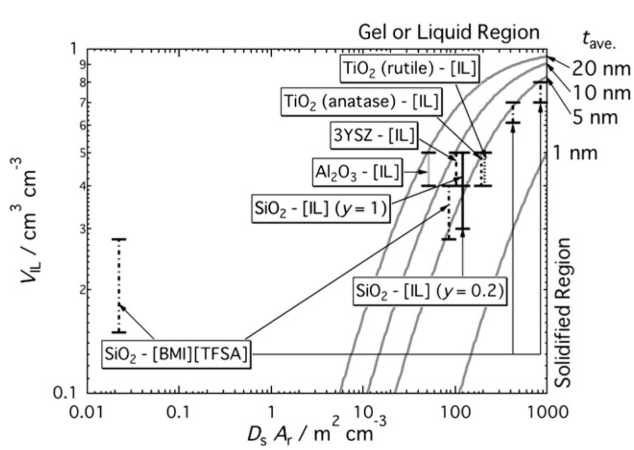

Fig. 4. State diagram of [yMLi$^+$][EMI$^+$][TFSA$^-$]–various oxide composites. The information regarding [BMI][TFSA]–SiO_2 composites from Ref. [4] are also provided.

3.2. Static- and MAS-NMR spectra

Fig. 5(a)–(c) compares the static ^{1}H, ^{19}F and ^{7}Li spectra, respectively, of ionic liquid bulk and typically SiO_2–40 vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] composites. The ionic liquid bulk had 5 peaks in the static ^{1}H NMR-spectrum, which could be assigned to be EMI-containing protons [21] (the numbers in the figure corresponds to the proton positions of EMI shown in Fig. 1). For SiO_2–40 vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] composites, the peaks were separated into four as in the figure because we could not distinguish peak-[3] from peak-[4]. Peaks for ^{7}Li and ^{19}F are considered to come from dissolved Li^+ and $TFSA^-$ containing fluorine, respectively.

According to Göbel et al., when the 1-ethyl-3-methylimidazlium triflate (EMI-TfO) is confined in the SiO_2 matrix, the chemical shifts of

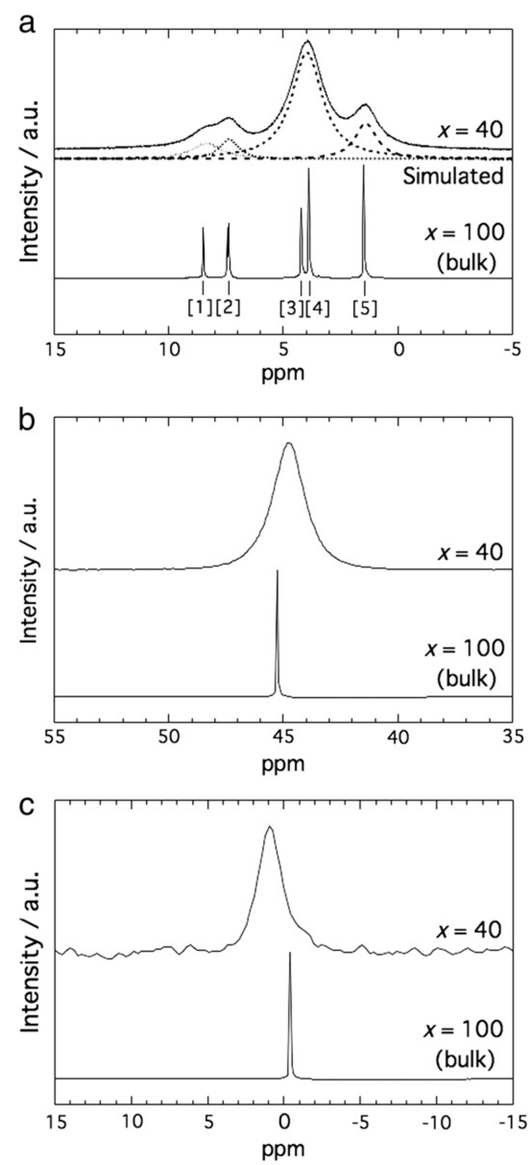

Fig. 5. Static (a) ^{1}H, (b) ^{19}F and (c) ^{7}Li NMR spectra of SiO_2–40 vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] and those of [1MLi$^+$][EMI$^+$][TFSA$^-$] bulk at room temperature.

peaks-[1] and -[2] shift to a lower field by 0.1 ppm and that of peak-[5] does to a higher field by 0.1 ppm. The other protons such as peaks-[3] and [4] did not show any changes [8]. Since the remarkable broadening of the spectra in confined H-containing species and small difference in chemical shifts, we could not quantitatively evaluate the change the individual peaks by quasi-solidification of ionic liquid. It was found that the chemical shifts of the peaks of ^{19}F and ^{7}Li changed from the bulk toward lower field by 0.5 ppm and higher field by 1.4 ppm, respectively. For Al_2O_3–40 vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] composites, the former nucleus did not move while the latter did to a higher field by 0.5 ppm, probably smaller changes in environments of F and Li than SiO_2–40 vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$]'s. It is clear that the individual spectra were broadened by the quasi-solidification of ionic liquid. This suggests that the ion dynamics is slowed by the quasi-solidification at the oxide particle surfaces similar to the cases of the preceding papers [17].

Fig. 6(a), (b) and (c) show MAS-NMR spectra of ^{1}H, ^{19}F and ^{7}Li, respectively, obtained for SiO_2–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] (x = 2.5–40) with a spinning rate of 10 kHz at room temperature. It was found from the ^{1}H MAS-NMR spectra that the composites have extra peaks at 4.4 ppm for SiO_2–[1MLi$^+$][EMI$^+$][TFSA$^-$], and 3.7 and 1.1 ppm for Al_2O_3–[1MLi$^+$][EMI$^+$][TFSA$^-$] as the volume ratio of ionic liquid decreases. Since the oxides without ionic liquid have peaks at those chemical shifts, the abovementioned peaks are essentially not originated from ionic liquid. Signal of the former is from the mixture of water molecules and silanol group and those of the latter are from water molecules and alminol group, respectively [24]. Actually, the area ratio of peak-[2] to peak-[1] in Fig. 6(a) keeps at 2 regardless of the x-values as expected from the ion structure. However, the ratio of peaks-[3] and [4] to peak-[1] exceeded 9 even when x = 40 although 5 is expected. The ratio increased by decreasing x, suggesting the remarkable contributions from water molecules and/or the silanol group.

In order to quantitatively evaluate the difference in HWHM (half-width at half-maximum), which is related to the motion of ionic species, by decreasing ionic liquid volume, the HWHM of ^{1}H, ^{19}F and ^{7}Li were evaluated. Since the contributions from water molecules or silanol/alminol groups could not be separately evaluated in static spectra, we evaluated the HWHM with ^{1}H MAS-NMR spectra of peak-[1] and peak-[2] for SiO_2 and Al_2O_3 composites, respectively. Fig. 7 shows the HWHM of ^{1}H, ^{19}F and ^{7}Li, which are normalized to those of x = 40. It was found that the HWHM gradually changes as a function of ionic liquid volume ratio, x, for both oxides. With decreasing ionic liquid volume ratios, the HWHM increased for H- and F-containing species for SiO_2 and all species for Al_2O_3. Significant difference was not observed for Li-containing species for SiO_2–ionic liquid composites. These may imply that the deviations of the dynamics of the ionic liquid from the bulk progress gradually with decreasing ionic liquid volume ratios. Details regarding the diffusivities of the constituent species will be discussed below.

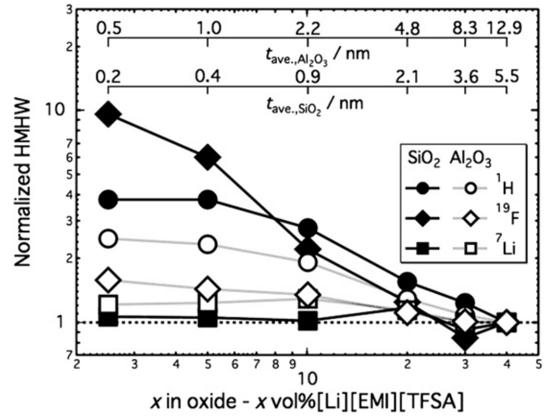

Fig. 7. HWHM of ^{1}H, ^{19}F and ^{7}Li to those normalized with x = 40 as a function of ionic liquid volume ratio for SiO_2– and Al_2O_3–[1MLi$^+$][EMI$^+$][TFSA$^-$] composites.

3.3. Apparent electrical conductivities

Fig. 8 shows the typical ac impedance spectra obtained for SiO_2–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] composites at 373–538 K. (a) and (b) are for x = 2.5 and 40, respectively. The apparent electrolyte resistance, $R_{app.}$, was evaluated as in the figure and converted into the apparent electrical conductivity, $\sigma_{app.}$, by,

$$\sigma_{app.} = \frac{1}{R_{app.}}\frac{L}{A} \tag{6}$$

where L and A are the thickness and the areas of the pressed body. The electrical conductivity of [1MLi$^+$][EMI$^+$][TFSA$^-$] is also evaluated by Eq. (6) by replacing $R_{app.}$ to bulk resistance, R. Fig. 9(a) and (b) shows the apparent electrical conductivity of SiO_2 and Al_2O_3–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] composites, respectively, as a function of inverse temperature. As expected in the preceding works [4–8], the ionic liquid quasi-solidified composites had liquid-like high apparent electrical conductivities such as $\log(\sigma/\mathrm{S\,cm^{-1}}) = -3.3$ to -2.0 and -3.4 to -2.1 for SiO_2– and Al_2O_3–40 vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] composites, respectively, in the temperature range of 323–538 K. It was also

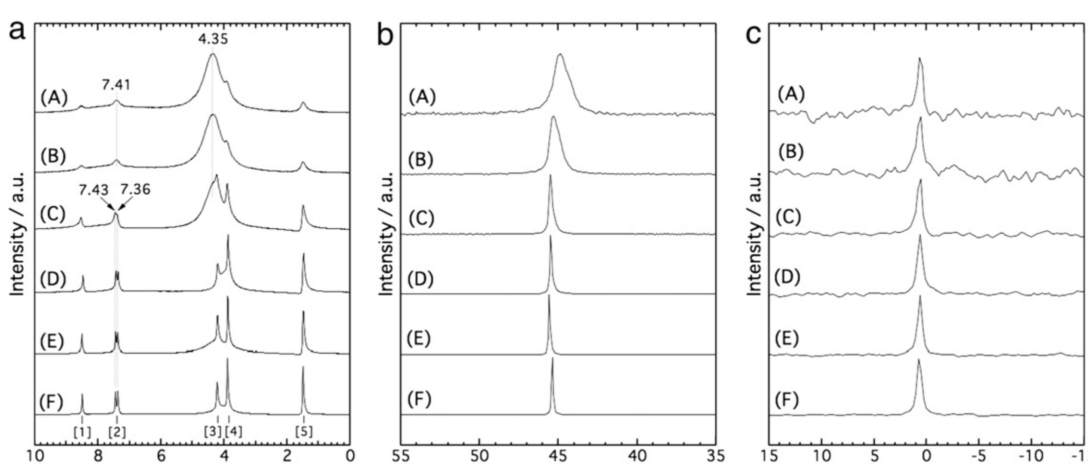

Fig. 6. (a) ^{1}H, (b) ^{19}F and (c) ^{7}Li MAS-NMR spectra of SiO_2–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] for (A) x = 2.5, (B) x = 5, (C) x = 10, (D) x = 20, (E) x = 30 and (F) x = 40 with a spinning rate of 10 kHz at room temperature.

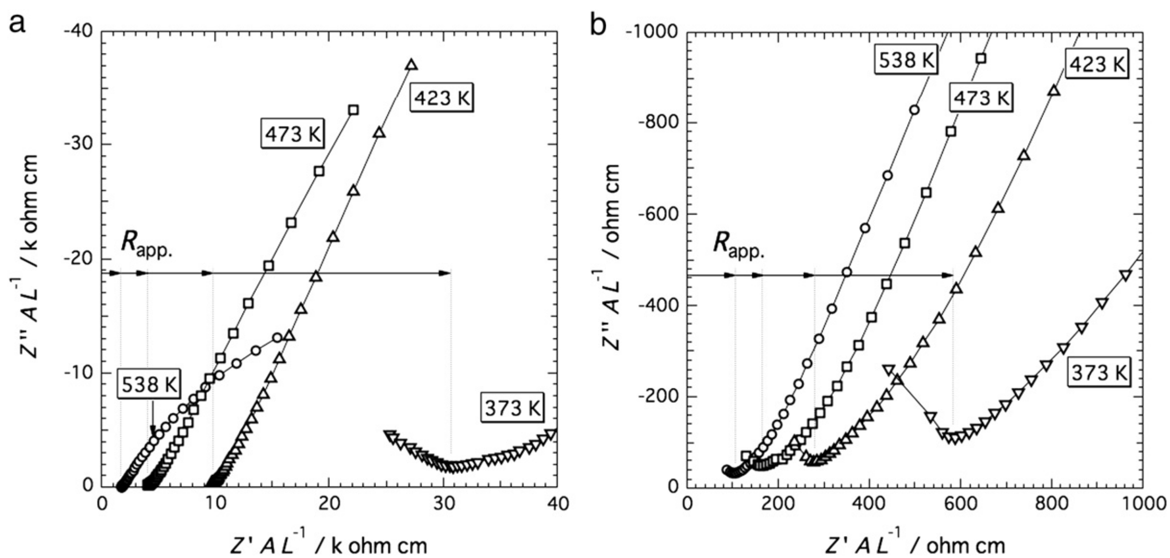

Fig. 8. Typical ac impedance spectra of SiO_2–x vol.%[$1MLi^+$][EMI^+][$TFSA^-$] composites for (a) $x = 0.025$ and (b) $x = 0.4$ in the temperature range of 373–538 K.

found that the conductivity monotonically increased by increasing the mixing ratio of ionic liquids for both composites, suggesting that the path for ionic species is essentially provided in the ionic liquid phase in the entire ionic liquid fraction examined here. It is worth noting that the composite has the considerable apparent conductivity even if the mixing ratio is small such as 2.5 vol.%. This implies that the ionic liquid is dispersed so as to allow the composite to percolate the conduction path across the compact as expected in the microstructural analyses.

The apparent activation energy of the electrical conductivity is shown in Fig. 10. Here, the apparent activation energy, $E_{app.}$, was derived by,

$$E_{app.} = -\frac{R}{F}\frac{\partial\left(\ln \sigma_{app.}T\right)}{\partial(1/T)}. \tag{7}$$

The term, $\partial(\ln \sigma_{app.}T)/\partial(1/T)$, was evaluated from the gradient of the neighboring plot points appearing in Fig. 9(a) and (b) by putting $\partial(\ln \sigma_{app.}T)/\partial(1/T) = \Delta(\ln \sigma_{app.}T)/\Delta(1/T) = (\ln \sigma_{n+1}T_{n+1} - \log \sigma_n T_n)/(1/T_{n+1} - 1/T_n)$ into Eq. (7), where temperature was defined by $\frac{1}{T} = \left(\frac{1}{T_{n+1}} + \frac{1}{T_n}\right)/2$. The electrical conductivity is known to obey non-Arrhenius-type temperature dependence, whereas we used Eq. (7) just for visualization of temperature dependence of composites as a function of ionic liquid volume ratio. The apparent activation energies as a function of temperature were shown in Fig. 10(a) and (b) for SiO_2 and Al_2O_3–x vol.%[$1MLi^+$][EMI^+][$TFSA^-$] composites, respectively. For ionic liquid bulk ($x = 100$), $\sigma_{app.}$ was replaced by the conductivity, σ. For SiO_2–x vol.%[$1MLi^+$][EMI^+][$TFSA^-$] composites, the apparent activation energy increased by decreasing the volume fraction of ionic liquid, which is similar to the behavior observed by Shimano et al. [4]. Similar tendency was also observed for SiO_2–[$0.2MLi^+$][EMI^+][$TFSA^-$] composites. On the other hand, for Al_2O_3–x vol.%[$1MLi^+$][EMI^+][$TFSA^-$] composites, the curves for $x = 5$–40 are duplicated for that of bulk. The gradient suddenly increased when $x = 0.025$.

Fig. 11(a), (b) and (c) shows the apparent electrical conductivities of x vol.%[$1MLi^+$][EMI^+][$TFSA^-$]–TiO_2 (rutile), TiO_2 (anatase) and 3YSZ, respectively. Similar to the cases of x vol.%[$1MLi^+$][EMI^+][$TFSA^-$]–SiO_2 or Al_2O_3 composites, the abovementioned composites exhibit liquid like

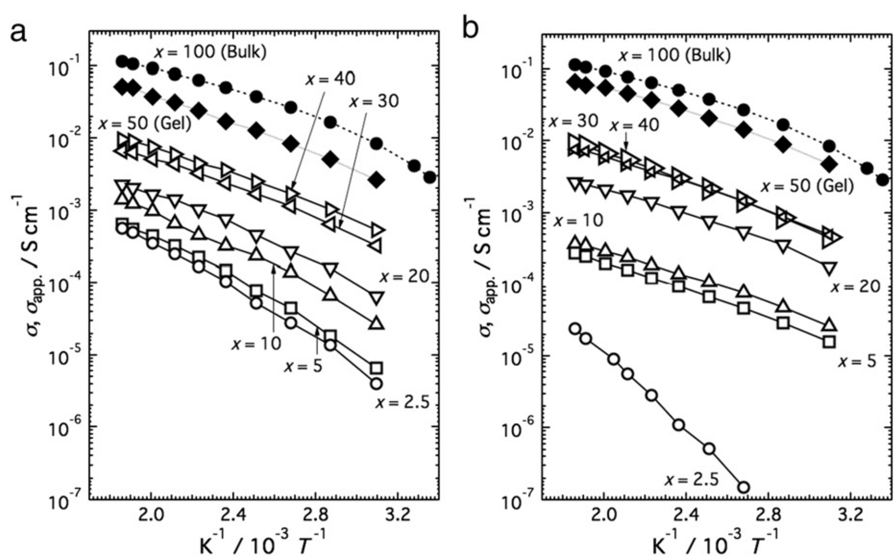

Fig. 9. The conductivity of [$1MLi^+$][EMI^+][$TFSA^-$] and the apparent conductivities of (a) SiO_2–x vol.%[$1MLi^+$][EMI^+][$TFSA^-$], (b) Al_2O_3–x vol.%[$1MLi^+$][EMI^+][$TFSA^-$] composites as functions of inverse temperature.

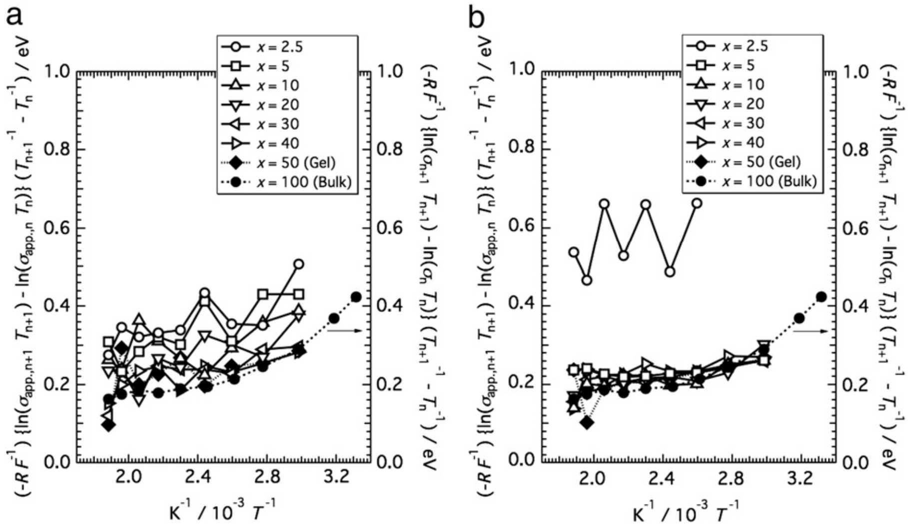

Fig. 10. The apparent activation energies of (a) SiO_2–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] and (b) Al_2O_3–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] as a function of inverse temperature.

high apparent conductivities such as $\log(\sigma/\mathrm{S\,cm^{-1}}) = -3.3$ to -2.0, -3.4 to -2.2 and -3.2 to 1.9, respectively, in the temperature range of 323–538 K with ionic liquid quasi-solidified composition, $x = 40$. The apparent electrical conductivity decreased with decreasing ionic liquid volume ratios, suggesting that the ionic liquid surely works as conduction paths. Fig. 11(A), (B) and (C) show the apparent activation energies, derived by Eq. (7), for x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$]–TiO_2 (rutile), TiO_2 (anatase) and 3YSZ, respectively. The apparent activation energies exhibit higher values than the bulk when x is below 5, 10 and 5 for TiO_2 (rutile), TiO_2 (anatase) and 3YSZ, respectively. This suggests that the dynamics of the ionic liquid of the composites deviates from the bulk with the above compositions.

As seen in Fig. 7, the HWHM of ^1H and ^{19}F MAS-NMR spectra increased gradually by decreasing the volume ratio of ionic liquid for both SiO_2 and Al_2O_3 even the composites having comparatively higher volume ratio of the ionic liquid. In such a situation, the deviation of the transport properties from the bulk is expected to gradually progress depending on ionic liquid volume ratio as seen in the electrical conductivity of SiO_2–ionic liquid composites. However, the apparent activation energy of Al_2O_3–ionic liquid suddenly increased when $x = 0.025$. This suggests that the diffusivities of the constituent species in the latter ionic liquid–oxide composites are indirectly related to the electrical conductivity probably due to the heterogeneity of coated ionic liquid thickness. Considering the temperature dependence of the apparent conductivity and NMR spectra, it might be considered that the conduction property in solidified ionic liquid layer essentially deviates from that in the bulk. The degree of the deviation might be more significant as quasi-solidified ionic liquid thickness decreases.

The Vogel–Tamman–Fulcher (VTF) formula [25,26], $\sigma = \sigma_0 exp\left(-\frac{B}{T-T_0}\right)$, is often employed to express the temperature dependence of the conductivity of ionic liquid having two [16,27] or more ionic species [16]. By combination with the Archie's law [28] to express the overall apparent electrical conductivity of the composites consisting of the oxide–liquid [4,29], $\sigma_{\mathrm{app.}} = aV_{\mathrm{IL}}^m\sigma$, we obtain $\sigma_{\mathrm{app.}} = aV_{\mathrm{IL}}^m\sigma_0 exp\left(-\frac{B}{T-T_0}\right)$. Regarding $aV_{\mathrm{IL}}^m\sigma_0$, B and T_0 as constants depending on the ionic liquid volume ratios, we analyzed their composition

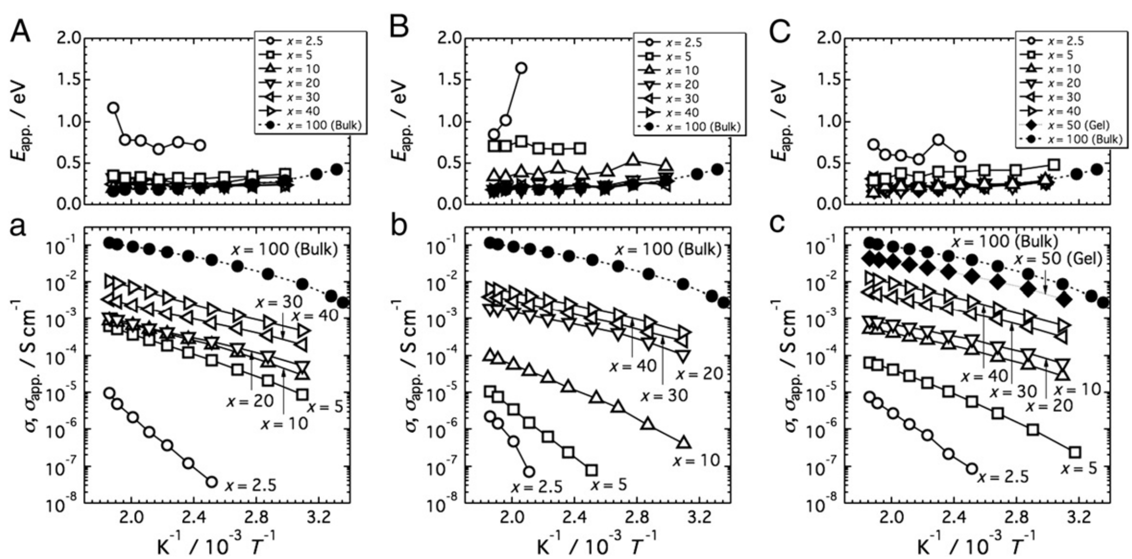

Fig. 11. The apparent conductivities of (a) TiO_2 (rutile)–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$], (b) TiO_2 (anatase)–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] and (c) 3YSZ–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] and the apparent activation energies of (A) TiO_2 (rutile)–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$], (B) TiO_2 (anatase)–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] and (C) 3YSZ–x vol.%[1MLi$^+$][EMI$^+$][TFSA$^-$] as functions of inverse temperature.

dependence. However, systematic trends of the ionic liquid volume ratio and/or the oxide species on the above parameters were not observed.

3.4. Dynamics of constituent species

As seen in Fig. 7, HWHM of ^{1}H, ^{19}F and ^{7}Li MAS-NMR spectra inferred the change of dynamics of ionic species by solidifications mixing together with oxide particles. As discussed in the previous chapter, the ionic liquid is dispersed at SiO_2 particles' surfaces higher than the other oxides'. By taking this into consideration, we investigated the self-diffusion coefficients of constituent species in SiO_2–[yMLi$^+$][EMI$^+$][TFSA$^-$] ($y = 0.2$ and 1) composites.

Fig. 12(a)–(c) shows the diffusion coefficients of ^{1}H, ^{19}F and ^{7}Li-containing species, D_H, D_F and D_{Li}, respectively, as a function of the ionic liquid volume ratio at 298–348 K. Diffusing species having the above nuclei are considered to be EMI$^+$ and EMI-TFSA for ^{1}H, TFSA$^-$, EMI-TFSA and [Li(TFSA)$_{n+1}$]$^{n-}$ [19,22,23] for ^{19}F, and [Li(TFSA)$_{n+1}$]$^{n-}$ for ^{7}Li [19]. Diffusion coefficients of ^{7}Li, ^{1}H and ^{19}F-containing species could be determined when the volume ratio of the ionic liquid is above 20% and 10%, respectively, for $y = 1$ while above 20% of all constituent species for $y = 0.2$. The PGSE-NMR signals could not be obtained on the composites with lower volume ratios than above. It is clear that the diffusion coefficients of all species decreased by the quasi-solidification at SiO_2 particle surfaces. The decrease of the diffusion coefficients for $y = 0.2$ was found to be more moderate than those for $y = 1$.

Table 2 summarizes the normalized diffusion coefficients of the ionic species in SiO_2–x vol.%[yMLi$^+$][EMI$^+$][TFSA$^-$] ($y = 0.2$ and 1) by those of the ionic liquid bulk as a function of ionic liquid volume ratio. For $y = 0.2$, with the changing state of ionic liquid from bulk, gel to quasi-solidified states, the diffusion coefficients of all species get smaller in the order. The difference tends to be more visible at lower temperatures. For $y = 1$, the diffusion coefficients of H and F-containing species become small as the volume ratio of ionic liquid decreased. The difference of the diffusion coefficient of Li-containing species is not remarkable within the composition we examined at all temperatures investigated. The differences in decreasing ratio of the diffusion coefficients of the ionic species in the quasi-solidified ionic liquid may infer the differences in association and/or solvation coordination from the bulk.

The partial conductivities of species i using self-diffusion coefficients of ionic species i, σ_i, is expressed by using Nernst–Einstein's equation,

$$\sigma_i = D_i \frac{n_i}{kT} e^2 \tag{8}$$

where i corresponds to EMI$^+$, TFSA$^-$ and Li$^+$ in the system we investigated. n_i, k, T and e are the number of ions per unit volume, Boltzmann constant, absolute temperature and the elementary charge, respectively. The electrical conductivity estimated by the self-diffusion coefficients are indirectly correlated to that evaluated by the electrochemical method, σ, because of interrelation among the ionic species [18,19,30–32] and oxide species as examined earlier in ion gel [18,19]. Since the diffusion coefficient of the above ionic species is unavailable only by the PGSE-NMR measurements, thus, we evaluated the interrelation of the diffusing species and the oxide surface by comparison of the diffusion coefficients of the constituent species as presented in Fig. 12.

The self-diffusion coefficient of species i is expressed by Stokes–Einstein's formula as,

$$D_i = \frac{kT}{c\pi\eta r_i} \tag{9}$$

where c and r_i are the constant and the Stokes radius of species i, respectively. Investigation of the ratios of the self-diffusion coefficients

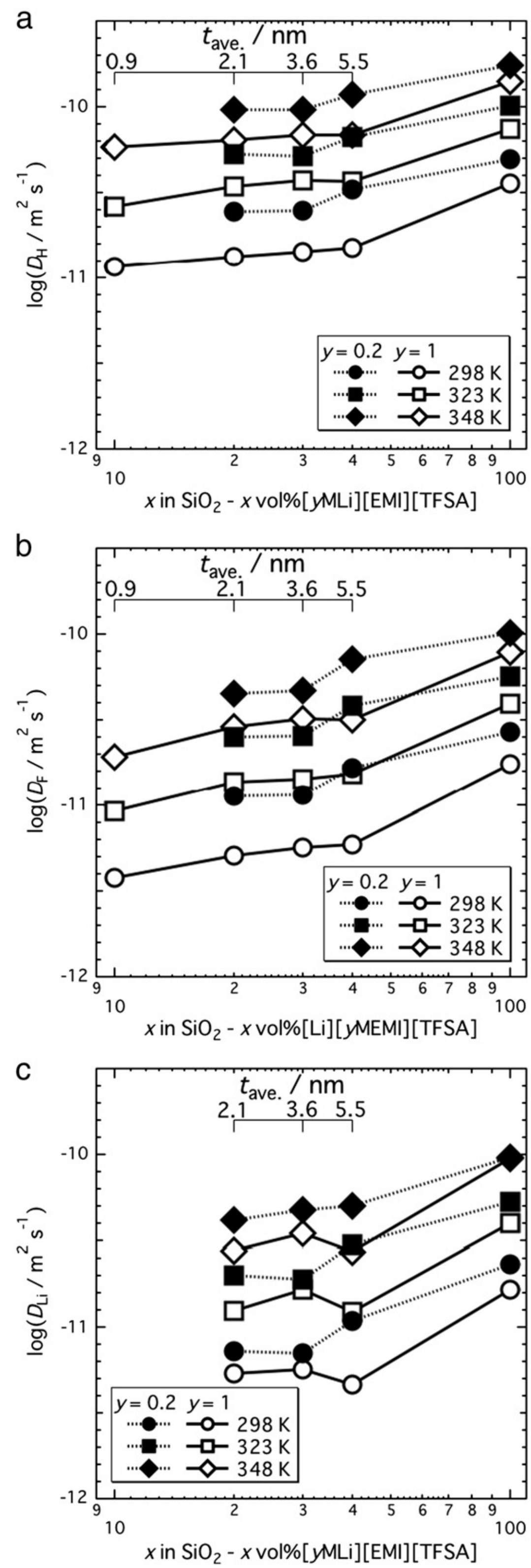

Fig. 12. Diffusion coefficients of (a) H-, (b) F- and (c) Li-containing species in SiO_2–x vol.%[yMLi$^+$][EMI$^+$][TFSA$^-$] composite as functions of x for $y = 0.2$ and 1 in the temperature range of 298–348 K.

such as D_H/D_F and D_{Li}/D_F is effective to imagine the ionization and solvation coordination. This is because the ratios reflect the ratios of the Stokes radius of the diffusing species. Fig. 13 shows the ratio of the diffusion coefficients of Li- and H- to that of F-containing species for

Table 2
Summary of the diffusion coefficients of constituent species in SiO_2–x vol.%[yMLi$^+$][EMI$^+$][TFSA$^-$] ($y = 0.2$ and 1) normalized by those with ionic liquid bulk ($x = 100$).

x	298 K			323 K			348 K		
	H	F	Li	H	F	Li	H	F	Li
$y = 0.2$									
40 (Gel)	0.66	0.61	0.47	0.66	0.67	0.57	0.67	0.70	0.52
30	0.50	0.42	0.30	0.51	0.45	0.36	0.55	0.46	0.50
20	0.49	0.42	0.31	0.52	0.44	0.38	0.55	0.44	0.43
$y = 1$									
40	0.42	0.34	0.28	0.50	0.39	0.31	0.49	0.41	0.29
30	0.40	0.32	0.34	0.50	0.36	0.42	0.49	0.41	0.37
20	0.37	0.29	0.32	0.46	0.35	0.31	0.46	0.37	0.29
10	0.33	0.22	–	0.35	0.24	–	0.41	0.25	–

SiO_2–x vol.%[yMLi$^+$][EMI$^+$][TFSA$^-$] as a function of the ionic liquid volume ratio at 298–348 K. It was found that the ratio, D_{Li}/D_F, remains constantly near unity regardless of the ionic liquid volume ratio and temperature for $y = 1$. The ratio for $y = 0.2$ tended to be smaller than that for $y = 1$ in the entire x and temperature ranges. In addition, as summarized in Table 2, the decreasing ratio of the normalized diffusion coefficients are similar between $y = 0.2$ and 1. On the other hand, the ratio, D_H/D_F, increases as the ionic liquid volume ratio decreases, and less remarkable temperature dependence was observed in the above ratios. As summarized in Table 2, the decreasing rate of the normalized diffusion coefficient of H-containing species is more moderate than that of F-containing species for both $y = 0.2$ and 1. Saito and her co-workers systematically investigated the ratios of the diffusion coefficients, D_H/D_F and D_{Li}/D_F, in Li-TFSA dissolved BMI-TFSA bulk as a function of Li-TFSA concentration. They suggested two possibilities of the increase and the decrease for the former and the latter ratios, respectively, by increasing Li-TFSA concentration. One is the change in temperature dependence of the diffusing species and the other is the ionization of the ion pair, [BMI$^+$][TFSA$^-$]. The ratio of the diffusion coefficient D_{Li}/D_F becomes unity when all TFSA$^-$ are solvated to Li$^+$ to form Li[(TFSA)$_{n+1}$]$^{n-}$ while the one becomes smaller than unity when the ratio of the concentration of TFSA$^-$ and/or EMI-TFSA is larger than that of Li$^+$ (or Li[(TFSA)$_{n+1}$]$^{n-}$) [19]. In the system we investigated, the ratio of the concentration of TFSA- to that of Li$^+$ is 5 and 21 for $y = 1$ and 0.2, respectively. Thus, the ratio, D_{Li}/D_F, of $y = 0.2$ is smaller than that of $y = 1$ because the free-TFSA$^-$ and/or EMI-TFSA of $y = 0.2$ is expected to be higher that of $y = 1$. As shown in Fig. 12, the ratios, D_H/D_F, of $y = 1$ are larger than those of $y = 0.2$. This may infer the progression of the ionization of the ion pair, [EMI$^+$][TFSA$^-$], by introduction of larger concentration of Li-TFSA also in the system we examined.

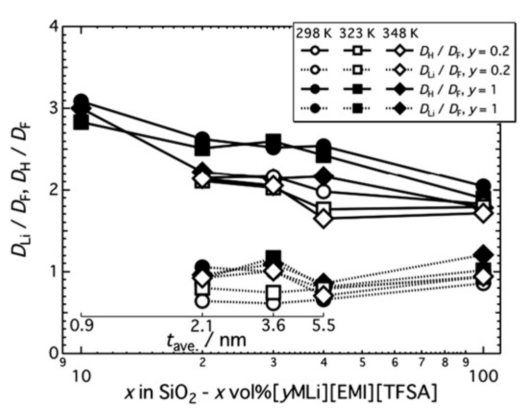

Fig. 13. Ratio of diffusion coefficients of Li- and H-containing species to that of F-containing species as a function of x in SiO_2–x vol.%[yMLi$^+$][EMI$^+$][TFSA$^-$] for $y = 0.2$ and 1 at 298, 323 and 348 K.

The ratios of the diffusion coefficients, D_H/D_F and D_{Li}/D_F, increases and remain constant, respectively, by decreasing the volume fraction of the ionic liquid, i.e. thickness of the quasi-solidified ionic liquid at SiO_2 particle surfaces in the quasi-solidified ionic liquid under constant y. When the Stokes radii of the diffusing species in the quasi-solidified ionic liquid are the same with those in the ionic liquid bulk, i.e. ionization and/or solvation coordination in the quasi-solidified ionic liquid is the same with those in the ionic liquid bulk, the decreasing ratios of the diffusion coefficients the diffusing species is expected to be common at a constant temperature. However, as emphasized in Table 2, the decreasing ratios of the quasi-solidified ionic liquid from the bulk depended on the diffusing species. Thus, it suggests that the ionization and/or solvation coordination in the quasi-solidified phase is not same with that in the bulk. Increase of D_H/D_F by decreasing the confined ionic liquid thickness might suggest the progression of the ionization of the ion pair, [EMI$^+$][TFSA$^-$], by the quasi-solidification for both compositions, resulting in the increase of the free-EMI$^+$ concentration. The concentration of free-EMI$^+$ for $y = 1$ is expected to be higher than that for $y = 0.2$. Insignificant dependence of D_{Li}/D_F on the ionic liquid thickness may suggest that the solvation coordination does not deviate from the bulk.

With comparatively high volume ratio of the ionic liquid such as $x = 40$, the dynamics of the ionic liquid is bulky regardless of the oxide compositions as shown in the apparent activation energies as in Figs. 10 and 11 even though the ionic liquid is quasi-solidified. Just beneath the oxide surface, the diffusion coefficients of the constituent species are expected to decrease because the effective viscosity, which is one of the parameters determining the diffusion coefficients as in Eq. (9), is enhanced with orders of magnitude [13]. Considering these, the enhancement of the effective viscosity and the change of the ionization coordination may be possible causes leading the deviation of the apparent activation energies of the composites from the bulk as in Figs. 10 and 11.

3.5. Apparent transport numbers

Eq. (10), combined with Eq. (8), is often employed to express the apparent transport number of species i [30],

$$t_i = \frac{\sigma_i}{\sum_i \sigma_i}. \quad (10)$$

Table 3 summarizes the apparent transport numbers of H, F and Li-containing species as a function of ionic liquid volume ratio and temperature. As shown in Table 3, the apparent transport number of H- and F-containing species increased and decreased, respectively, by the quasi-solidification of ionic liquid. This is considered to be due

Table 3
Summary of the transport numbers of constituent species, t_i (i = H, F and Li), for SiO_2–[yMLi$^+$][EMI$^+$][TFSA$^-$] composites ($y = 0.2$ and 1).

x	298 K			323 K			348 K		
	H	F	Li	H	F	Li	H	F	Li
$y = 0.2$									
100	0.626	0.359	0.015	0.619	0.364	0.017	0.610	0.373	0.017
40 (Gel)	0.646	0.343	0.011	0.617	0.369	0.014	0.602	0.385	0.013
30	0.667	0.323	0.010	0.651	0.337	0.012	0.649	0.334	0.017
20	0.664	0.326	0.010	0.658	0.329	0.013	0.660	0.325	0.015
$y = 1$									
100	0.576	0.355	0.069	0.554	0.369	0.077	0.532	0.375	0.093
40	0.634	0.315	0.051	0.623	0.323	0.054	0.594	0.345	0.061
30	0.624	0.312	0.064	0.625	0.302	0.073	0.581	0.342	0.077
20	0.631	0.303	0.066	0.627	0.314	0.059	0.576	0.338	0.086

to the increased concentration of EMI^+ by disassociation of ion pair, [EMI^+][$TFSA^-$], as discussed in the previous chapter. The difference of the transport number of Li-containing species is negligible because the molar concentration of Li^+ is small against those of the other ionic species even if the diffusion coefficient of Li-containing species decreased more than other species.

4. Conclusions

Quasi-solid-state lithium ion conducting composites consisting of ionic liquid and various oxides are prepared and the ion transport properties are evaluated by ac impedance and NMR spectrometry. The thickness of the confinable ionic liquid was evaluated to be approximately 5–10 nm regardless of the oxide species. As a result, we have successfully synthesized the ionic liquid–oxide composites having the apparent conductivities of, typically $10^{-3.3}$–$10^{-2.0}$ S cm^{-1} of SiO_2–ionic liquid composites, in the temperature range of 323–538 K. Based on the temperature dependence of the electrical conductivity and the diffusion coefficients of the ionic species suggested that the ion dynamics of the quasi-solidified ionic liquid deviates from those of the bulk. The diffusion coefficients of all constituent species decreased compared to those in the bulk for SiO_2–ionic liquid composites. D_H/D_F increased in the quasi-solidified ionic liquid while D_{Li}/D_F kept similar to the bulk. These suggest that SiO_2 surface promotes disassociation of [EMI^+][$TFSA^-$] while there is no significant effect on solvation coordination of $[Li(TFSA)_{n+1}]^{n-}$. Although the diffusion coefficient of Li^+ decreased by the quasi-solidification of ionic liquid, the transport number of Li-containing species was not remarkably decreased due to small concentration of Li^+. It exhibits 0.051–0.093 for $y=1$ and 0.010–0.017 for $y=0.2$ in the temperature range of 298–348 K regardless of the ionic liquid volume ratio.

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